Itô Calculus

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1 Setting

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions and supporting a standard Brownian motion $W = (W_t)_{t\geq 0}$ adapted to $(\mathcal{F}_t)$. Sample paths of W are almost surely continuous and nowhere differentiable, with unbounded total variation on every compact interval.

Definition 1 (Quadratic variation). *For a stochastic process $X = (X_t)$ and a partition $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ write $\|\Pi\| = \max_i(t_i - t_{i-1})$. The* quadratic variation *of X on $[0, t]$ is*

$$[X]_t \;:=\; \lim_{\|\Pi\|\to 0} \sum_{i=1}^{n} \big(X_{t_i} - X_{t_{i-1}}\big)^2,$$

provided the limit exists in probability. For Brownian motion, $[W]_t = t$ a.s.

Definition 2 (Itô integral). *Let $H = (H_t)$ be a progressively measurable process with $\mathbb{E}\int_0^T H_s^2\,ds < \infty$. The* Itô integral *of H against W is the L^2-limit*

$$\int_0^t H_s\,dW_s \;:=\; \lim_{\|\Pi\|\to 0} \sum_{i=1}^{n} H_{t_{i-1}}\big(W_{t_i} - W_{t_{i-1}}\big).$$

Evaluation at the left endpoint *t_{i-1} is essential: it makes $t \mapsto \int_0^t H\,dW$ a continuous square-integrable martingale, with Itô isometry $\mathbb{E}\big[(\int_0^t H\,dW)^2\big] = \mathbb{E}\int_0^t H_s^2\,ds$.*

Definition 3 (Itô process). *A process X is an* Itô process *if*

$$X_t = X_0 + \int_0^t b_s\,ds + \int_0^t \sigma_s\,dW_s,$$

for adapted b, σ with $\int_0^T |b_s|\,ds < \infty$ and $\int_0^T \sigma_s^2\,ds < \infty$ a.s. We write this symbolically as $dX_t = b_t\,dt + \sigma_t\,dW_t$.

2 Itô's Lemma

Theorem 4 (Itô's lemma, 1D). *Let $X_t = X_0 + \int_0^t b_s\,ds + \int_0^t \sigma_s\,dW_s$ be an Itô process and let $f \in C^2(\mathbb{R})$. Then $f(X_t)$ is again an Itô process and*

$$df(X_t) \;=\; f'(X_t)\,dX_t + \tfrac{1}{2}\,f''(X_t)\,\sigma_t^2\,dt.$$

Equivalently, in integral form,

$$f(X_t) - f(X_0) \;=\; \int_0^t f'(X_s)\,b_s\,ds + \int_0^t f'(X_s)\,\sigma_s\,dW_s + \tfrac{1}{2}\int_0^t f''(X_s)\,\sigma_s^2\,ds.$$

Proof sketch via second-order Taylor expansion and $(dW_t)^2 = dt$. Fix a partition $\Pi = \{0 = t_0 < \cdots < t_n = t\}$ and write $\Delta_i X = X_{t_i} - X_{t_{i-1}}$, $\Delta_i t = t_i - t_{i-1}$, $\Delta_i W = W_{t_i} - W_{t_{i-1}}$. By telescoping and a second-order Taylor expansion of f around $X_{t_{i-1}}$,

$$\begin{aligned} f(X_t) - f(X_0) &= \sum_{i=1}^{n} \Big[f(X_{t_i}) - f(X_{t_{i-1}}) \Big] \\ &= \sum_i f'(X_{t_{i-1}})\, \Delta_i X + \tfrac{1}{2} \sum_i f''(X_{t_{i-1}})\, (\Delta_i X)^2 + R_\Pi, \end{aligned}$$

where R_Π collects the higher-order Taylor remainder, controlled by the modulus of continuity of f'' on a compact set containing the path of X.

First-order term. Since $\Delta_i X = \int_{t_{i-1}}^{t_i} b_s\, ds + \int_{t_{i-1}}^{t_i} \sigma_s\, dW_s$, the sum $\sum_i f'(X_{t_{i-1}})\, \Delta_i X$ converges in L^2 to

$$\int_0^t f'(X_s)\, b_s\, ds + \int_0^t f'(X_s)\, \sigma_s\, dW_s,$$

by definition of the Itô integral and dominated convergence; the left-endpoint evaluation $f'(X_{t_{i-1}})$ is exactly what the Itô construction requires.

Second-order term — the key identity $(dW_t)^2 = dt$. Expand

$$(\Delta_i X)^2 = b_{t_{i-1}}^2 (\Delta_i t)^2 + 2 b_{t_{i-1}} \sigma_{t_{i-1}} (\Delta_i t)(\Delta_i W) + \sigma_{t_{i-1}}^2 (\Delta_i W)^2 + o(\Delta_i t)\text{-corrections.}$$

The $(\Delta_i t)^2$ contributions sum to $O(\|\Pi\| \cdot t) \to 0$. The cross-terms $(\Delta_i t)(\Delta_i W)$ have L^2-norm $O(\Delta_i t^{3/2})$, summing to $O(\|\Pi\|^{1/2}) \to 0$. The surviving piece is

$$\sum_i f''(X_{t_{i-1}})\, \sigma_{t_{i-1}}^2\, (\Delta_i W)^2.$$

Because $\mathbb{E}[(\Delta_i W)^2] = \Delta_i t$ and $\mathrm{Var}[(\Delta_i W)^2] = 2(\Delta_i t)^2$, a direct L^2 computation shows that $(\Delta_i W)^2$ may be replaced by $\Delta_i t$ in the limit:

$$\sum_i f''(X_{t_{i-1}})\, \sigma_{t_{i-1}}^2 (\Delta_i W)^2 \xrightarrow[\|\Pi\| \to 0]{L^2} \int_0^t f''(X_s)\, \sigma_s^2\, ds.$$

This is precisely the heuristic identity $(dW_t)^2 = dt$ made rigorous, and it is an instance of the quadratic variation $[W]_t = t$.

Remainder. $R_\Pi \to 0$ in probability because f'' is continuous and the path of X is bounded on $[0, t]$ almost surely. Collecting the surviving pieces yields the formula stated. $\square$

Remark 5 (Why the left endpoint?)**.** *The proof shows the limit of* $\sum H_{t_{i-1}} \Delta_i W$ *depends on the choice of evaluation point. Replacing* $H_{t_{i-1}}$ *with the midpoint* $H_{(t_{i-1}+t_i)/2}$ *gives the* Stratonovich *integral, which obeys the classical chain rule* $df(X_t) = f'(X_t) \circ dX_t$ *but is not a martingale. The right-endpoint choice peeks at the future and produces a different, anticipating object. Itô's left-endpoint choice gives a martingale, at the cost of the second-order correction term in Itô's lemma.*

Remark 6 (Canonical example)**.** *With* $f(x) = x^2$ *and* $X_t = W_t$*, Itô's lemma yields*

$$d(W_t^2) = 2 W_t\, dW_t + dt \quad \Longrightarrow \quad \int_0^t W_s\, dW_s = \tfrac{1}{2}(W_t^2 - t).$$

The classical guess $\frac{1}{2} W_t^2$ *is off by exactly the Itô correction* $-t/2$.